V-766-APDE MIDTERM EXAM

18 January 2026 (1200 hours) - 21 January 2026 (1200 hours) Icelandic Time

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Maximum Possible: 100 marks

Instructions:

- Answer ALL 5 Questions
- Show ALL working (i.e. **do not use CUSTOMIZED FUNCTIONS**).
  - Q1 (10 marks),
  - Q2 (20 marks),
  - Q3 (25 marks),
  - Q4 (25 marks),
  - Q5 (20 marks),
- Answer all the questions on the spreadsheet. Label each tab in the spreadsheet (e.g. Solution to Q2), solution to Q3) etc.)
- When filling up green/yellow coloured cells on the spreadsheet for Question 1, please note that green coloured cells refer to the fact that only values are required while yellow coloured cells imply that a formula is required.
- Please load up the midterm solutions on the portal when done and email me a copy at kravindran@annuitysystems.com and ravindran@ru.is.
- While discussion is permitted between students on how to approach the problem, no sharing of solutions or copying or looking into another's solutions or forwarding someone else's solution is allowed.
- Please be mindful of the deadline which is **21 January 2026 (1200 hours)** Icelandic Time
- You are allowed to use any of the spreadsheet tabs I have used in class, copy it over and make the necessary modifications to arrive at your solution.
- Please email me if you have any questions or need further clarifications. I will reply to questions via announcement boards. NO questions on lectures or exercises would be answered during this period.

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1) **Valuing Vanilla European-Style Option:** Using the Black-Scholes equations, and assuming that the current price of an Apple share is \$255, life of option is 8 months, annualized volatility is 30%, the 8-month discount factor is $e^{-0.04}$ and annualized continuously compounded dividend rate is 4%.

   a. Value the following options by filling in the colored cells on the Tab "Question 1"

      i. At-the-money forward put option
      ii. At-the-money spot call option

   b. Using the results of a. and the put-call parity, find the values of

      i. At the money forward call option
      ii. At the money spot put option

2) **Valuing Vanilla European-Style Option With Truncated Payoff:** An investor wants to pay an upfront premium (at the inception of the contract), $Option_{van\ trunc\ upfront}$, to

acquire an European-style option with the following payoff at maturity:

$$\begin{cases} 0 & if\ S_T < X_1 \\ X_2 - S_T & if\ X_1 \le S_T \le X_2 \\ S_T - X_2 & if\ X_2 \le S_T \le X_3 \\ 0 & S_T > X_3 \end{cases}$$

where S_T represents the stock price on option maturity (i.e. time T) and $0 < X_1 < X_2 < X_3$.

a) Using the idea of replication, find the value of $Option_{van\ trunc\ upfront}$ when $\sigma = 30\%, r = 4\%, q = 1\%, X_1 = 80, X_2 = 100, X_3 = 120, T = 1$ and $S_0 = 110$

b) To check your answer in a), find the value of $Option_{van\ trunc\ upfront}$ using a 30-step tree[1].

c) Find the value of Z (the conditional premium paid at time T only if the stock ends up being in the interval X_1 to X_3 at time T) using the numerical values given in a), if the profit to the purchaser of the option at time T is now given by the following:

$$\begin{cases} 0 & if\ S_T < X_1 \\ X_2 - S_T - Z & if\ X_1 \le S_T \le X_2 \\ S_T - X_2 - Z & if\ X_2 \le S_T \le X_3 \\ 0 & S_T > X_3 \end{cases}$$

3) **Valuing a best choice option:** An investor wants to pay an upfront premium, $Option_{van\ best\ choice}$, to acquire an European-style call option with the following payoff at maturity:

$$\max(X_1 - S_T, 0, S_T - X_2)$$

where S_T represents the stock price on option maturity (i.e. time T), X_1 and X_2 are option strikes (where $0 < X_1 < X_2$) and the option holder is allowed to choose between a call and a put on the choice time/option maturity ($= T$)

a. Find the expression for $Option_{van\ best\ choice}$ that can be used to value the option.

b. Find the value of the option using the expression in a., when $\sigma = 30\%, r = 5\%, q = 2\%, X_1 = 90, X_2 = 110, T = 0.5$ and $S_0 = 100$

c. Find the value of the option when $\sigma = 30\%, r = 5\%, q = 2\%, X_1 = 90, X_2 = 110, S_0 = 100$ AND when the choice is made at time 0.2 years where the choice made at this time allows the option owner to choose between a call option (life 0.3 years, strike 110) and put option (life 0.3 years, strike 90).[2]

[1] You need to extend the tree we did in class (i.e. $N = 6$ to $N = 30$).

[2] To solve this problem, you may want to use the 30-step binomial tree that you created in Q2, to similarly create a tree for the time period 0 – 0.2 years and then combine the ability to choose at time 0.2 years between a European-style call and put options both with lives 0.3 years and respective strikes of 110/90 respectively.

4) **Option Premium, Prob of Exercise, Sensitivity:** Assume that for a non-dividend paying stock

- an European style at-the-money spot put (with maturity T) has a 30% probability of being exercised,
- sensitivity of the European style at-the-money spot call (with maturity T) to changes in underlying stock price is 0.65
- \$10 today would be worth \$12 at time T in the future

Given the above information, find the values of

a. current stock price if the European style at the money spot call premium is \$2
b. European-style at-the-money spot binary call option with a life of $2T$ years and a bet of \$100.

5) Question To Be Given Out on Tuesday (20 Jan 2026) at 1200 hours: